Project Report

Project Title: IntelliGrade AI
Academic Performance Prediction and Recommendation System

Submitted To:
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Artificial Intelligence

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IntelliGrade AI

Academic Performance Prediction and Recommendation System

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1. Introduction

Academic performance is one of the most important indicators of student success. Educational institutions continuously seek methods to monitor student progress and identify potential academic risks before final examinations. Traditional assessment methods often provide feedback after examinations have already taken place, limiting opportunities for early intervention.

To address this issue, IntelliGrade AI was developed as an Academic Performance Prediction and Recommendation System. The system utilizes Machine Learning techniques to analyze student academic data and predict future performance. It also calculates GPA, identifies risk levels, and provides personalized recommendations to help students improve their academic outcomes.

2. Project Overview

IntelliGrade AI is a web-based intelligent system that combines Machine Learning and Web Development technologies to assist students in monitoring and improving their academic performance.

The system allows students to:

- Enter personal academic information.
- Manage up to five courses.
- Predict final examination performance using AI.
- Calculate course percentages and GPA.
- Determine academic grades.
- Identify academic risk levels.
- Receive personalized study recommendations.
- Visualize results through an interactive dashboard.

System Workflow

1. Student enters personal information.
2. Student enters course performance data.
3. Machine Learning model predicts final examination score.
4. Academic calculation engine computes percentage, GPA, and grades.
5. Dashboard displays analytical results and recommendations.\

3. Problem Statement

Many students struggle to evaluate their academic standing during a semester. Most academic evaluations occur only after final examinations, making it difficult for students to identify weaknesses and improve performance in time.

The absence of predictive academic analysis can result in poor grades, reduced GPA, and academic probation risks. Therefore, an intelligent system capable of predicting academic performance and providing early recommendations is required.

4. Objectives

The objectives of IntelliGrade AI are:

- Predict student academic performance using Machine Learning.
- Calculate GPA automatically.
- Generate academic grades.
- Analyze attendance and risk levels.
- Provide personalized recommendations.
- Improve student awareness regarding academic progress.
- Support informed academic decision-making.

5. Research and Information Gathering

Before development, extensive research was conducted regarding academic performance prediction systems, educational analytics, and Machine Learning algorithms.

The Student Performance Dataset was selected from Kaggle after evaluating multiple educational datasets. Research was also conducted on grading systems, GPA calculations, attendance analysis, and predictive modeling techniques.

Dataset title & Description

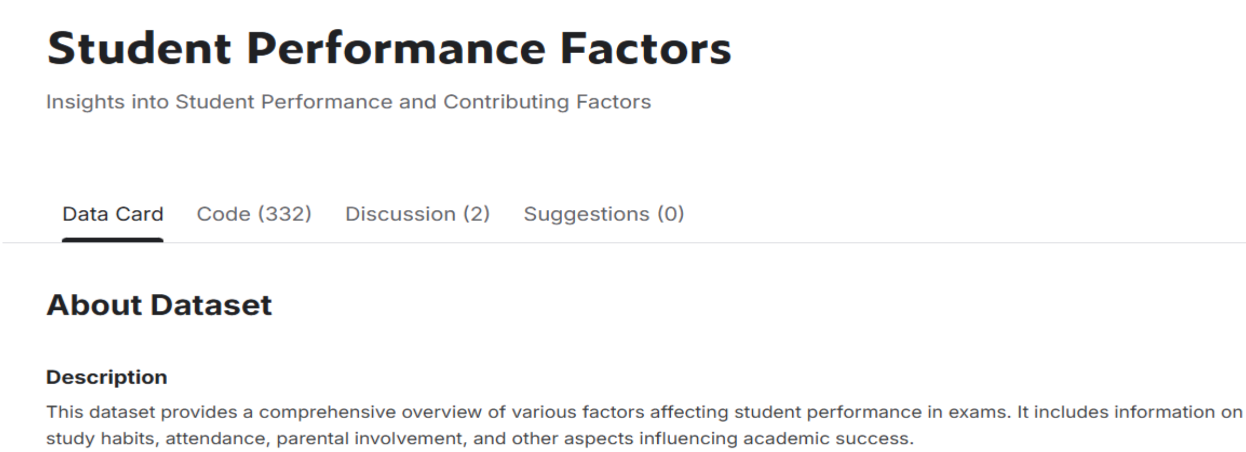

Figure 1:Dataset Title & Description

6. Tools and Technologies Used

Technology	Purpose
Python	Core Programming Language
Flask	Backend Framework
HTML5	User Interface Structure
CSS3	Styling and Layout
JavaScript	Frontend Interactivity
Pandas	Data Processing
Scikit-Learn	Machine Learning
Joblib	Model Saving and Loading
Visual Studio Code	Development Environment
Kaggle	Dataset Source

7. Dataset Description

The project utilizes the Student Performance Dataset obtained from Kaggle.

Dataset Information

- Total Records: 6,607
- Total Features: 20
- Target Variable: Exam Score

Important Features

- Hours Studied
- Attendance
- Previous Scores
- Tutoring Sessions
- Sleep Hours
- Family Income
- Access to Resources
- Motivation Level
- Teacher Quality
- Physical Activity

The dataset contains educational, social, and behavioral factors that influence academic performance.

The column description is given below;

Column Descriptions

Attribute	Description
Hours_Studied	Number of hours spent studying per week.
Attendance	Percentage of classes attended.
Parental_Involvement	Level of parental involvement in the student's education (Low, Medium, High).
Access_to_Resources	Availability of educational resources (Low, Medium, High).
Extracurricular_Activities	Participation in extracurricular activities (Yes, No).
Sleep_Hours	Average number of hours of sleep per night.
Previous_Scores	Scores from previous exams.
Motivation_Level	Student's level of motivation (Low, Medium, High).
Internet_Access	Availability of internet access (Yes, No).
Tutoring_Sessions	Number of tutoring sessions attended per month.
Family_Income	Family income level (Low, Medium, High).
Teacher_Quality	Quality of the teachers (Low, Medium, High).
School_Type	Type of school attended (Public, Private).
Peer_Influence	Influence of peers on academic performance (Positive, Neutral, Negative).
Physical_Activity	Average number of hours of physical activity per week.
Learning_Disabilities	Presence of learning disabilities (Yes, No).
Parental_Education_Level	Highest education level of parents (High School, College, Postgraduate).

Distance_from_Home	Distance from home to school (Near, Moderate, Far).
Gender	Gender of the student (Male, Female).
Exam_Score	Final exam score.

Figure 2:Column Description

8. Methodology

8.1 Data Collection

The Student Performance Dataset was collected from Kaggle.

8.2 Data Preprocessing

Data preprocessing involved:

- Handling missing values
- Feature selection
- Data transformation
- Data formatting

8.3 Model Training

A Random Forest Regression model was trained using selected features from the dataset.

8.4 Model Evaluation

The model was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

8.5 Model Deployment

The trained model was saved using Joblib and integrated into the Flask application.

Output:

```
Dataset Shape: (6607, 20)

===== MODEL PERFORMANCE =====
MAE: 1.23
RMSE: 2.46
R² Score: 0.6118

===== TOP IMPORTANT FEATURES =====
                  Feature  Importance
1              Attendance    0.396890
0           Hours_Studied    0.258373
6         Previous_Scores    0.088307
9       Tutoring_Sessions    0.034045
3     Access_to_Resources    0.027480
14      Physical_Activity    0.025622
2    Parental_Involvement    0.023972
5             Sleep_Hours    0.022812
10          Family_Income    0.020948
13         Peer_Influence    0.015287

Model Saved Successfully!
Files Generated:
1. intelligrade_model.pkl
2. columns.pkl
```

Figure 3:Output After model training

9. Machine Learning Model

9.1 Model Evaluation Metrics

This project did not use Decision Tree. The final model used was Random Forest Regression. The model performance was evaluated using the following metrics:

- Mean Absolute Error (MAE): Measures the average absolute difference between the predicted values and the actual values.
- Root Mean Squared Error (RMSE): Measures the square root of the average squared differences between the predicted values and the actual values.
- R-squared (R^2) Score: Also called the coefficient of determination, it shows how well the model explains the variation in the target variable.

Output:

```
Dataset Shape: (6607, 20)

===== MODEL PERFORMANCE =====
MAE: 1.23
RMSE: 2.46
R² Score: 0.6118
```

Figure 4:Model Evaluation Metrics

9.2 Random Forest Regression

Random Forest Regression is an ensemble learning technique that combines multiple Decision Trees to improve prediction accuracy.

Advantages:

- Higher accuracy
- Reduced overfitting
- Better generalization
- Reliable predictions

9.3 Reason for Selecting Random Forest

Random Forest was selected because it provided better performance and more stable predictions compared to individual Decision Trees.

10. System Implementation

10.1 Frontend Development

The frontend was developed using HTML, CSS, and JavaScript.

Pages developed include:

- Landing Page
- Student Information Form
- Course Information Form
- Dashboard

Landing Page:

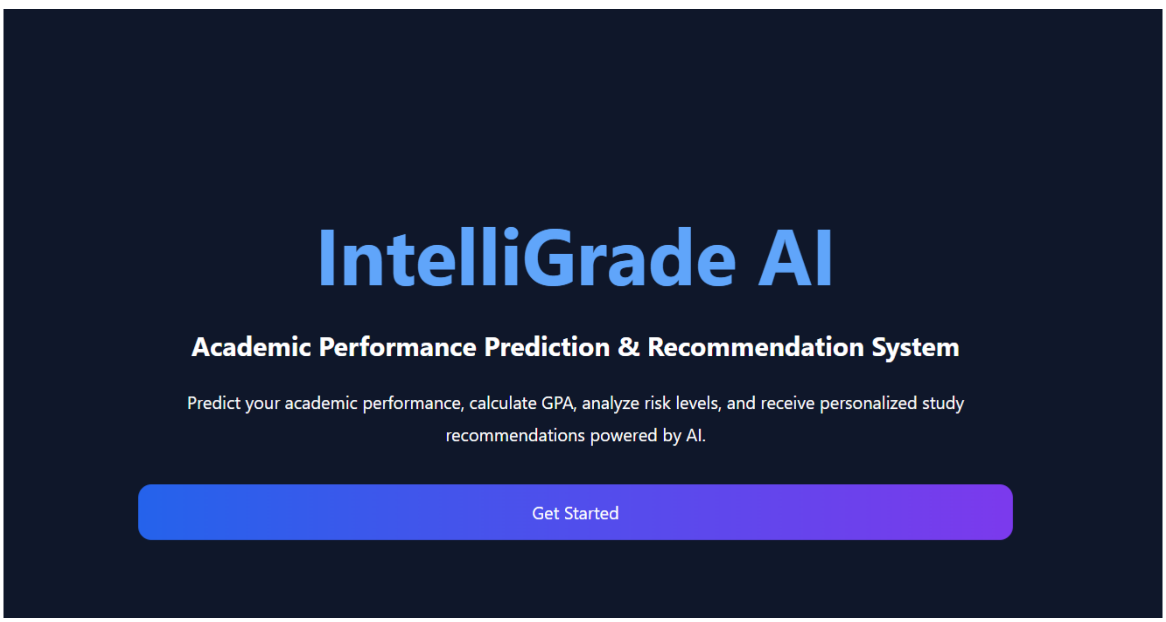

Figure 5:Dashboard

Student Dashboard:

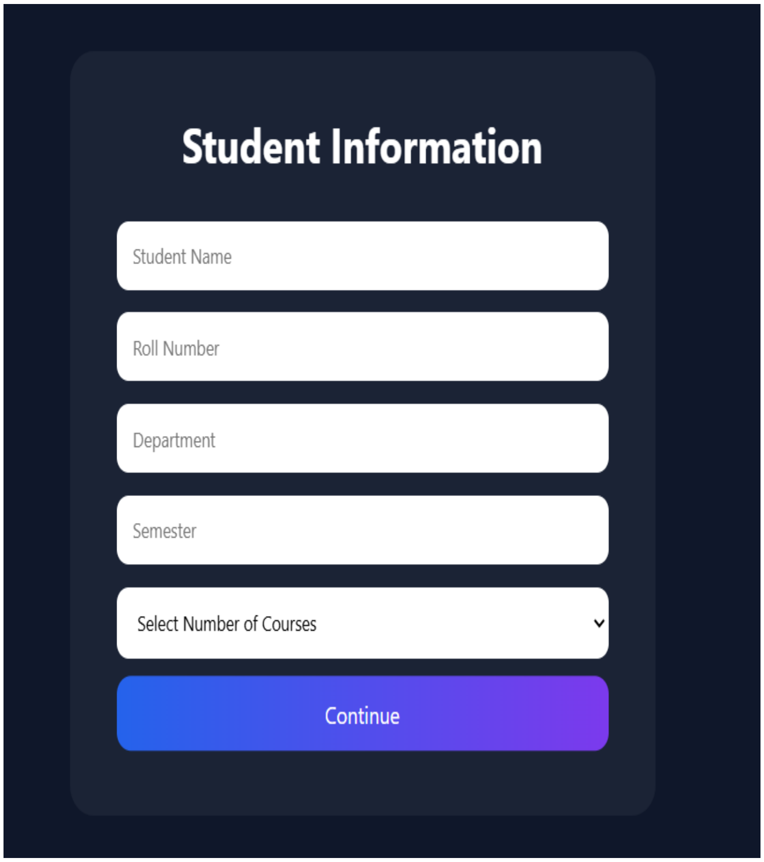

Figure 6:Student Dashboard

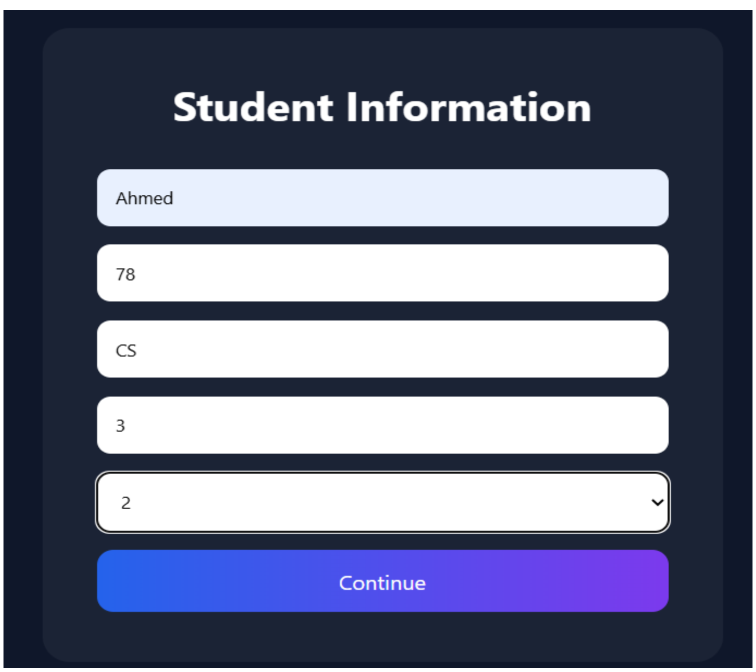

Figure 7:Student Information Portal

Courses Details

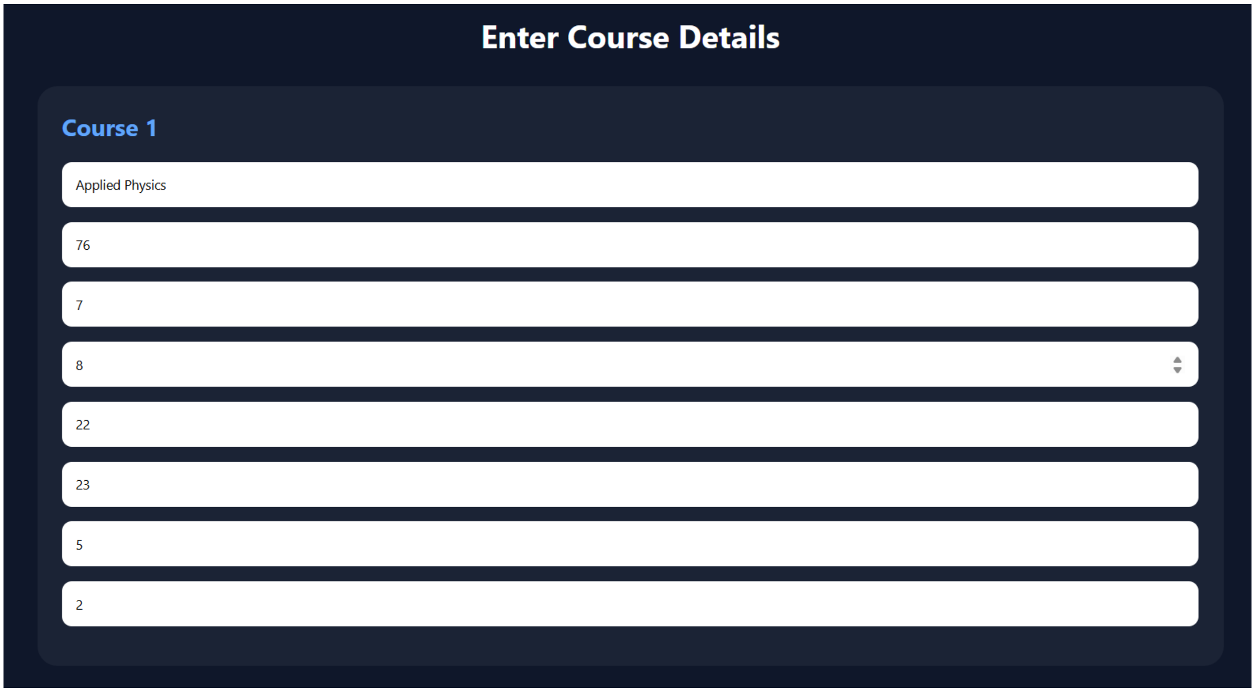

Figure 8:Courses Information System

Figure 9:Courses Information System

Prediction Dashboard

Figure 10:Prediction Dashboard

Figure 11:Recommendation Dashboard

10.2 Backend Development

The backend was developed using Flask.

Major functionalities include:

- Route Management
- Session Handling
- Machine Learning Integration
- GPA Calculation
- Grade Calculation

11.Conclusion

IntelliGrade AI is an Academic Performance Prediction and Recommendation System developed using Machine Learning and Web Technologies. The system predicts students' final exam performance, calculates GPA, determines academic risk levels, and provides personalized recommendations based on academic data such as attendance, quizzes, assignments, and study hours. The project successfully demonstrates the application of Artificial Intelligence in education by helping students monitor and improve their academic performance through data-driven insights. Using a Random Forest Regression model and an interactive Flask-based web interface, the system provides accurate predictions and meaningful analysis in a user-friendly manner. Overall, IntelliGrade AI achieves its objectives effectively and highlights the potential of AI-powered solutions in enhancing academic decision-making and student success.

12. Future Enhancements

Future improvements may include:

- Database integration
- User authentication system
- Real-time analytics
- Advanced performance visualizations
- Mobile application support
- Faculty monitoring dashboard
- Cloud deployment

13. References

i. Kaggle Student Performance Dataset.

ii. Scikit-Learn Documentation.

iii. Flask Documentation.

iv. Python Official Documentation.

v. Pandas Documentation.

vi. Machine Learning Research Articles.